

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q14-0381

Front Royal 138 kV
November 29, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0381. The Interconnection Customer proposes a 49.3 MW solar photovoltaic facility interconnecting at the Front Royal 138 kV substation in the Virginia area. The request was received on September 20, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 228 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,415,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

North Mesa – Opequon 138.0 kV: 118.6% loading (263 MVA) following the outage of the Kearneyville – Clark County 345.0 kV line; DFAX = 9.3%.

For the loss of Kearneyville – Clark County 345.0 kV Line, the Lcec No. 5 – Opequon 138.0 kV loads to 118.2% of its 256 MVA rating. Project DFAX: 9.3%.

The Wheeler – Pasco Cnty Solid Waste Resource Recovery 138.0 kV is overloaded to 115.5% of its 384 MVA emergency rating for the loss of the Arcola – Twin Lakes Alte 138.0 kV. The project contributes 7.7% DFAX (3.8 MW) to this overload.

For the loss of Artesia Country Club – Air Force Base 345.0 kV, the G-P Gypsum – Brighton 138.0 kV loads to 115.1% of its 381 MVA rating. Project DFAX: 5.0%.

The Ocean Road – Monolith 138.0 kV is overloaded to 114.3% of its 239 MVA emergency rating for the loss of the International Paper Valliant Ok – Kearneyville 345.0 kV. The project contributes 8.8% DFAX (4.3 MW) to this overload.

For the loss of International Paper Valliant Ok – Kearneyville Substation 345.0 kV, the University Of Maine – Three Peaks 138.0 kV loads to 112.4% of its 227 MVA rating. Project DFAX: 8.8%.

The Plum Pt Energy Sta – University Of Maine 138.0 kV is overloaded to 111.9% of its 222 MVA emergency rating for the loss of the International Paper Valliant Ok – Kearneyville 345.0 kV. The project contributes 8.8% DFAX (4.3 MW) to this overload.

The Ocean Rd – Three Peaks 138.0 kV is overloaded to 111.5% of its 232 MVA emergency rating for the loss of the International Paper Valliant Ok – Kearneyville 345.0 kV. The project contributes 8.8% DFAX (4.3 MW) to this overload.

For the loss of Kearneyville – International Paper Valliant Ok 345.0 kV, the Plum Pt Energy Sta – Mabon 138.0 kV L loads to 110.5% of its 212 MVA rating. Project DFAX: 8.8%.

The Mabton – Southampton 138.0 kV L is overloaded to 109.8% of its 208 MVA emergency rating for the loss of the International Paper Valliant Ok – Kearneyville 345.0 kV Line. The project contributes 8.8% DFAX (4.3 MW) to this overload.

Celeron – Southampton 138.0 kV: 109.5% loading (200 MVA) following the outage of the International Paper Valliant Ok – Kearneyville 345.0 kV; DFAX = 8.8%.

E Ellensburg – Terror Lake Microgrid 138.0 kV Line: 108.4% loading (237 MVA) following the outage of the International Paper Valliant Ok – Kearneyville 345.0 kV; DFAX = 6.8%.

For the loss of Air Force Base – Artesia Country Club 345.0 kV line, the Artesiacountry Club 138 kV/1 kV Transformer loads to 107.2% of its 465 MVA rating. Project DFAX: 9.5%.

Dixleta – Rustic 345.0 kV: 106.7% loading (1,314 MVA) following the outage of the Dixleta – Hughson 345.0 kV; DFAX = 12.3%.

For the loss of International Paper Valliant Ok – Kearneyville 345.0 kV, the W Hills – E Ellensburg 138.0 kV line loads to 105.8% of its 231 MVA rating. Project DFAX: 6.8%.

For the loss of Artesia Country Club – Air Force Base 345.0 kV, the Ross Fork Substation – Artesia Country Club 138.0 kV loads to 104.3% of its 529 MVA rating. Project DFAX: 9.3%.

For the loss of Mcconnico – Huntley 138.0 kV, the Belleair – Old Chapel 138.0 kV loads to 101.2% of its 265 MVA rating. Project DFAX: 8.6%.

Tombstone – Marcus Hook Energy Lp 500.0 kV L: 100.1% loading (2,341 MVA) following the outage of the Curecanti – Moser Generating Sta 345.0 kV; DFAX = 6.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
North Mesa – Opequon 138.0 kV	Rebuild line with higher-capacity conductor	\$3,509,000	8% (\$267,000)
Lcec No. 5 – Opequon 138.0 kV	Rebuild line with higher-capacity conductor	\$5,061,000	8% (\$385,000)
Wheeler – Pasco Cnty Solid Waste Resource Recovery 138.0 kV	Rebuild line with higher-capacity conductor	\$5,073,000	2% (\$83,000)
G-P Gypsum – Brighton 138.0 kV	Rebuild line with higher-capacity conductor	\$7,700,000	10% (\$749,000)
Ocean Road – Monolith 138.0 kV	Rebuild line with higher-capacity conductor	\$745,000	6% (\$42,000)
University Of Maine – Three Peaks 138.0 kV	Rebuild line with higher-capacity conductor	\$2,489,000	6% (\$140,000)
Plum Pt Energy Sta – University Of Maine 138.0 kV	Rebuild line with higher-capacity conductor	\$1,625,000	6% (\$91,000)
Ocean Rd – Three Peaks 138.0 kV	Rebuild line with higher-capacity conductor	\$6,788,000	6% (\$381,000)
Plum Pt Energy Sta – Mabon 138.0 kV L	Rebuild line with higher-capacity conductor	\$2,808,000	6% (\$157,000)
Southampton – Mabton Substation 138.0 kV	Reconductor line	\$1,323,000	6% (\$74,000)
Celeron – Southampton 138.0 kV	Reconductor line	\$766,000	6% (\$43,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E Ellensburg – Terror Lake Microgrid 138.0 kV Line	Reconductor line	\$1,999,000	7% (\$134,000)
Artesia Country Club 138-1 kV TX	Replace transformer with higher-rated unit	\$5,232,000	9% (\$449,000)
Rustic – Dixleta 345.0 kV Line	Reconductor line	\$33,489,000	0% (\$79,000)
W Hills – E Ellensburg 138.0 kV line	Reconductor line	\$1,537,000	7% (\$103,000)
Ross Fork – Artesia Country Club 138.0 kV L	Reconductor line	\$1,916,000	6% (\$109,000)
Belleair – Old Chapel 138.0 kV	Reconductor line	\$202,000	6% (\$12,000)
Tombstone – Marcus Hook Energy Lp 500.0 kV	Reconductor line	\$39,634,000	0% (\$117,000)
	Total allocated to project		\$3,415,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.